

Hey, everybody. Welcome to your chapter 16 review. If you've been following along in order with the chapters, you know that this episode will focus on core training concepts. This is a deeper dive into another component of integrated training.

Core training isn't just about getting that beach-ready bod. A stable core is actually critical for improving posture, enhancing performance, increasing resistance to injury, and accelerating injury rehabilitation. A properly designed core training program can be a key component of an overall training plan, and used to achieve a broad range of goals. The basic objective of core training is the development of core stability, endurance, strength, and power.

It's inevitable that as a fitness professional, you're going to hear a lot of core related goals, whether it's for strength or aesthetic reasons. After this chapter, you should be able to identify the physical benefits of core training, differentiate between the core stabilization and movement systems, describe the mechanisms of the drawing in maneuver and abdominal bracing, identify proper progressions of core training exercises, and employ methods to effectively execute, instruct, and cue core training exercises.

All right, now let's get to the core of core training. Core stability and endurance refer to the ability to maintain proper spinal and hip posture while the extremities are moving. Core strength refers to the ability to contract and control movements of the torso in all directions, meaning flexion, extension, and rotation, whereas core power involves explosive movements of the core musculature.

The core is defined by the structures that make up the lumbopelvic hip complex, or LPHC. It includes local and global core musculature that stabilize the spine while producing efficient arm and leg movements. Your local core muscles generally attach on or near the vertebrae.

Local muscles such as the rotators, multifidus, transverse abdominis, and diaphragm provide dynamic control of the spinal segments, limiting excessive compression, shear, and rotational forces between spinal segments. Global core muscles are positioned more superficially on the trunk in comparison to local muscles. Global muscles, like the rectus abdominus, external and internal abdominal obliques, and erector spinae, act to move the trunk, transfer loads between the upper and lower extremities, and provide stability of the spine by stabilizing multiple segments together as functional units.

You should make sure that local muscles are effective in stabilizing the spine when you're introducing more advanced core exercises that utilize global muscles to move the trunk. It's also really important to ensure both local and global core muscles demonstrate adequate endurance because of the continuous need for optimal spinal stability and precise trunk movements during many core exercise activities.

When you're designing a core training program, the local and global muscles should both be trained in order to develop proper core stability and overall movement efficiency. When you think about it, the core has as much to do with optimizing posture as it does your midsection. Core stabilization, endurance, and strength is imperative for maintaining the natural curvatures of the spine, both at rest and during movement.

Large curvatures of the spine away from midline are considered abnormal and may be considered either structural or functional scoliosis. Core muscle strengthening has even been found to improve spinal alignment in individuals with functional scoliosis. It's important to be aware of those possible abnormalities and know how to work with individuals who have these conditions.

Core muscles help to protect the spine from harmful forces during functional activities. As mentioned, effective core training focuses on optimizing core stability, endurance, strength, and power. Focusing on each of these components is an important part of ensuring individuals can effectively stabilize their core and forcefully move their trunk for optimal performance.

Core training has also been demonstrated to improve injury resistance. How? Well, core muscle training has been found to improve balance, and is important for optimizing lower extremity biomechanics that may decrease the risk of knee injury. Low back pain, or LBP, for short, is another common cause of pain, discomfort, and even disability, that can be improved with core stabilization exercises.

Again, if you think about how and where core muscles attach and how they function, it makes perfect sense. Just remember, however, to stay within your scope of practice and work closely with health care providers if needed, who are trained to diagnose and treat injuries or perform the functions of a rehabilitation specialist.

I want to spend a little time on a couple specific strategies to enhance the function and ability of the local and global core muscles to stabilize the LPHC. First, the drawing in maneuver, it's often used as an effective exercise to increase pelvic stabilization and transverse abdominis activation. It's critical for activating local core muscles.

To perform the drawing in maneuver, pull in the region just below the navel toward the spine, and maintain the cervical spine in a neutral position to improve posture, muscle balance, and stabilization. Next, activating the global core muscles using the bracing technique is also beneficial. Bracing is commonly referred to as a bearing down or tightening of the global muscles by consciously contracting them.

Research has shown that muscular endurance of global and local musculature, when contracted together, create the most benefit for those with low back pain compared with traditional low back pain training methods. Bracing focuses on global trunk stability, not just segmented vertebral stability, meaning that when given the proper endurance training, global muscles will also work to stabilize the spine.

Now, while the drawing in maneuver and the bracing technique are critical for activating core musculature and helping increase a person's awareness of their core muscle engagement, there are many, many exercises outlined in the chapter that can be effective for improving clients' core stability endurance, strength, and power. There are many variables that can also be manipulated when designing a core training program, including planes of motion, ranges of motion, speed of motion, volume, and exercise modalities.

Just be cautious not to change too many variables at one time when you're progressing clients through their exercise program, in order to ensure they're able to demonstrate appropriate mastery at each stage. You also need to keep an eye on quality of movement so that you aren't allowing sacrifices in form just to feel like there's progress happening.

So how do you design a core training program with proper progressions? Start with core exercises that involve little motion of the spine and target the local core musculature, such as marching, floor bridges, Ball cobras, Planks, Side planks, Dead bugs, and pallof or anti-rotation presses, to name a few. The next progression should incorporate more motion at the spine and hips and target global core muscles. A few examples are the floor crunch, back extension, Reverse Crunch, Knee up cable rotation, and Cable chops.

The last core exercise progression involves explosive movement through the trunk and extremities, and includes exercises such as Medicine ball chest passes, Pullover throws, Soccer throws, Wood chop throws, and Overhead throws. For core training, there is an endless variety of additional exercises not listed that you can utilize within client training programs.

As with any training method, form and technique must be mastered to ensure proper muscle activation and to avoid injury. Keep in mind the progression of core exercises, starting from little motion of the spine, increasing motion through the spine and hips, and lastly, explosive movements involving the trunk, hips, and extremities. As with any exercise, make sure to monitor the five kinetic chain checkpoints carefully.

Overall, it's important for a comprehensive core training program to be systematic, progressive, functional, and incorporate a variety of isometric, concentric, and eccentric muscle actions. When you're developing a core training program, emphasize increasing proprioceptive demand initially to make things more challenging for clients, instead of simply increasing the external resistance. And I'll say it again. Quality over quantity, you especially want to emphasize quality of movement across the lumbopelvic hip complex.

Bottom line here is that core training is critical for improving posture, enhancing performance, increasing injury resistance, and accelerating injury rehabilitation. If you get a six pack in the process, that's just icing on the cake. The real value of enhancing core stability and strength is helping prepare your clients to be able to better handle the physical demands of training, exercise, and life.

And for your clients with more athletic goals, there are so many ways to build programs to help them reach their optimal performance levels in a progressive and effective way. So have fun developing those core workouts. They're well worth it. All right, until we chat again, be well, and happy studying.